

Python surface

`compose()` `MeasurementData` `fit()` / `fit_fast()`
user-facing model-construction API

`FitGraph` `FitOptions` `FitResult`
Pydantic schema mirror — validated at the boundary

request JSON

result JSON

PyO3 wheel boundary

every `#[pyfunction]` crosses as a JSON string — no shared memory layout
Python↔Rust schema parity enforced at commit time, not at run time

Rust workspace — 11 crates

spectrafit-core

PyO3 binding crate — serialises requests and results

spectrafit-solver

structure-routed dispatch — picks the solver family from problem structure

levenberg-
marquardt

dogleg

newton-cg

varpro

separable NLS

spectrafit-trust-region

faer-native trust-region core

graph · models

fit graph · 32 kernels

spectrafit-types

shared type foundation — the one crate every other crate depends on

arrows show dependency direction; only the spine is drawn

spectrafit-builder — compile-time exhaustiveness gate; an unhandled model variant fails to build, not at run time